BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: MRD-XII-21 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** MRD-XII-21 (A) VERSION: C

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e. A,B,C, or D.

Q1. ..... is a condition of large volume urine produced.

(A) Polyphagia
(B) Polydipsia
(C) Polyuria
(D) Polyphyrobia

Q2. Pain receptors are called .....

(A) Pacinians
(B) Nociceptors
(C) Tactile receptor
(D) Mechano receptor

Q3. Which one regulates the bone response to stress and mechanical load?

(A) Osteoblast
(B) Osteocytes
(C) Osteoclast
(D) All of these

Q4. ADH is secreted by the ..... lobe of pituitary gland.

(A) Anterior
(B) Posterior
(C) Both A & B
(D) None of these

Q5. Which one is not a chemical mutagen?

(A) Formaldehyde
(B) Acridine
(C) X-Rays
(D) Mustard gas

Q6. Kidney chronic failure occurs due to .....

(A) Antibodies
(B) Hypertension
(C) Alcohol
(D) Toxins

Q7. Which one is the symptom of pneumonia?

(A) Very high fever
(B) Rapid pulse rate
(C) Severe chest pain
(D) All of these

Q8. Epididymis length is about ..... feet.

(A) 12
(B) 15
(C) 17
(D) 20

Q9. Identical twins results from the fertilization of .....

(A) One ovum by one sperm
(B) One ovum by two sperms
(C) Two ova by two sperms
(D) Two ova by one sperm

Q10. Which one is stop codon?

(A) CAU
(B) UAG
(C) UAA
(D) Both B & C

Q11. Which one is the salient feature of Darwin-Wallace theory?

(A) Variation
(B) Overproduction
(C) Struggle for existence
(D) All of these

Q12. Which one bacterium is Aerobic?

(A) Azobactor
(B) Clostridium
(C) Azotobacter
(D) None of these

Q13. Cellular rearrangements and migrations occur during ..... stage.

(A) Morula
(B) Blastula
(C) Ampulla
(D) Gastrulation

Q14. A sarcomere is the region of a myofibril between two successive ..... line.

(A) M
(B) I
(C) H
(D) Z

Q15. In denaturation step of PCR, the template is heated at .....

(A) 54 °C
(B) 60 °C
(C) 70 °C
(D) [illegible in source scan]

Q16. Biogas contains ..... methane.

(A) 20−30%
(B) 30−40%
(C) 50−75%
(D) 80−90%

Q17. In Griffith experiment, mice developed pneumonia when they were injected with .....

(A) R-type bacteria
(B) Heat killed S-type bacteria
(C) Heat killed R-type bases
(D) None of these

Q18. Advantage of pecking orders to .....

(A) Avoids injury to the stronger animals
(B) Protect territory
(C) Find suitable mate
(D) None of these

Section-B

Time Allowed: 2:40 Hours | **Maximum Marks:** 40

Q2. Write short answers to any ten (10) of the following parts. All parts carry equal marks.

(i) What are the causes and risk factors of Tuberculosis?

(ii) Differentiate between osmoregulators and osmoconformers.

(iii) Define Hinge joints and Ball Socket joints.

(iv) Discuss Prostate Gland.

(v) Draw a label diagram of the Pancreas showing Islets of Langerhans.

(vi) Write down four applications of PCR.

(vii) What is the role of Microbes in Alcohol industry?

(viii) Differentiate between Primary Succession and Secondary Succession.

(ix) Write a short note on Endosymbiosis?

(x) What do you know about the semi conservative model of DNA?

(xi) Discuss sex-related traits with an example.

(xii) What is meant by co-dominance?

(xiii) Draw a label diagram of transverse section (T.S) of the kidney.

Section-C

Maximum Marks: 27

> Note: Answer any three (3) questions. All questions carry equal marks.

Q3. Define xerosere. Discuss various stages of xerosere on lithosere.

Q4. How Mendel performed dihybrid cross? Explain with example.

Q5. Discuss in detail the mechanism of PCR reaction.

Q6. Write notes on any two of the following.

(a) What are the smooth muscles?

(b) Types of mutation.

(c) Role of microbes in pharmaceutical industry.